

Online Appendix to

“Surprises, Shapes or Machines? Forecasting the Market’s Response to Macroeconomic Releases with Dynamic Time Warping (DTW)”

Benjamin Steel, Jack Duncan, Aaryen Mehta, Reza Zamani and Hoshea Zeng

OA.1 Release families traded by market

Table OA.1 lists the release families admitted by the impact screen in at least one year from 2018 to 2026 (rank in the screen at the start of 2026 in parentheses). CPI is always traded.

Table OA.1: Release families admitted by the impact screen (top ten in at least one year, 2018–2026).

ES	NQ	ZN	YM
CPI (1), Employment Report (2), Fed interest on reserves (3), FOMC decision (4), ISM Manufacturing (5), U. of Michigan (6), ISM Services (7), ISM Services Prices Paid (8), JOLTS (9), Case–Shiller NSA (10), Retail Sales, S&P Global Manufacturing PMI, GDP, Trade Balance, Philadelphia Fed, IBD/TIPP	CPI (1), Employment Report (2), Fed interest on reserves (3), FOMC decision (4), Dallas Fed (5), U. of Michigan (6), ISM Manufacturing (7), ISM Services (8), ISM Services Prices Paid (9), PPI (10), ISM Milwaukee, Retail Sales, S&P Global PMIs, Conference Board, Trade Balance, GDP	ISM Services Prices Paid (1), Employment Report (2), FOMC decision (3), CPI (4), ISM New Orders (5), ISM Manufacturing (6), ISM Services (7), Fed interest on reserves (8), S&P Global PMIs (9, 10), Retail Sales, Factory Orders, FHFA, U. of Michigan, Trade Balance	Employment Report (1), Fed interest on reserves (2), CPI (3), FOMC decision (4), ISM Manufacturing (5), Retail Sales (6), ISM Services (7), JOLTS (8), PPI (9), Trade Balance (10), Business Inventories, U. of Michigan, GDP, Wholesale Inventories, Wholesale Trade Sales, FHFA, S&P Global Services PMI, Pending Home Sales, Empire Manufacturing, Case–Shiller

OA.2 Parameters of every approach

Table OA.2: Fixed parameters. None is tuned on the test years.

Approach	Parameters
Surprises	family overlap 80%; at least 24 earlier releases for z ; cap ± 5 ; top 10 families by impact (plus CPI); at least 24 releases for a line sign; CPI model threshold = $1 \times$ round-trip cost
DTW ladder	path 31 minutes ending at entry, z-scored; Sakoe–Chiba band 5 minutes; lookback 3 years; at most 600 candidates; $k = 15$; weights $1/d \times$ 2-year half-life; gate: outside the 30th–70th percentiles; size median(σ)/ σ , cap 2; windows at $T+1, \dots, T+121$
DTW event games	anchors $T-60, \dots, T+60$; 60-minute path sampled every 3 minutes; pool Jobless Claims, JOLTS, PCE; lookback 2 years; DTW window 5 steps; one contract
Classifier	walk-forward; features from surprises, price paths and momentum; 30-minute horizon; positions in $[-1, 1]$
Portfolios	weights $\propto 1/\sigma_i$, daily net P&L at $\frac{1}{4}$ tick, 2018–2020

OA.3 Additional results

No state-conditioned model significantly beats the surprise-only model in any market: the change in Sharpe ratio ranges from -0.21 to $+0.08$ in ES, from -0.05 to $+0.03$ in NQ and from -0.20 to $+0.17$ in YM, where no difference is significant (smallest bootstrap $p = 0.23$). Adding PCE and NFP to CPI dilutes the signal in ES (surprise-only Sharpe 0.99, 0.71 and 0.59) but much less in YM (0.84, 0.89 and 0.78).

OA.4 Reproducibility map

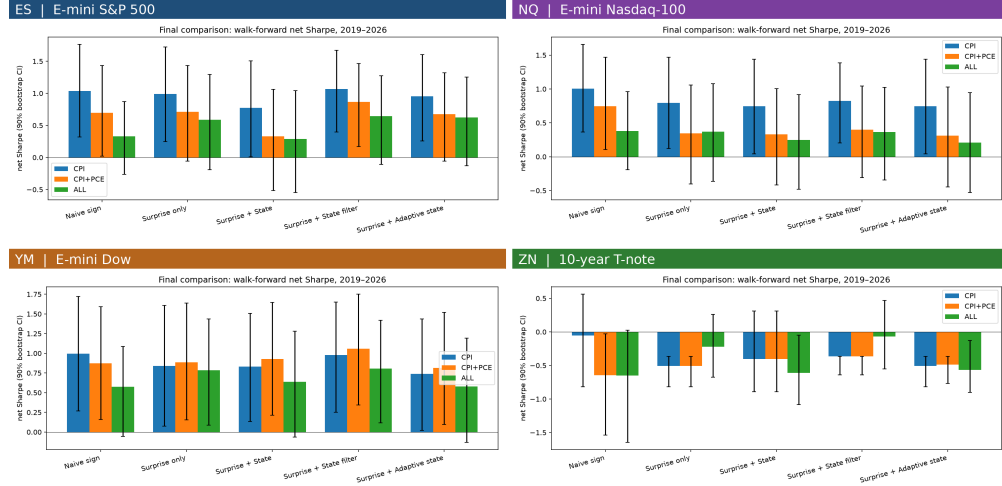


Figure OA.1: Net Sharpe ratio of the CPI strategies (naive, surprise model and three state-conditioned models) in three release universes (CPI; CPI + PCE; CPI + PCE + NFP), ES, NQ, YM and ZN, 2019–2026, at 2 ticks + \$4.50.

Table OA.3: Programs that produce each exhibit (repository <https://github.com/jack-duncan/blackrock-intraday>).

Exhibit	Program
Figures 1–3, Table 3	Reza/notebooks/02_event_response_analysis, 02b
Figure 4	Reza/notebooks/03_state_conditioning, 03b
Sections V.A–B and VII, Figures 6, 9–10	Reza/notebooks/04–06 and b versions
Table 5	Reza/notebooks/08_team_comparison_equal_costs
Figure 7	combined/surprise_dtw_book
Tables 4 and 6, Figure 8	combined/team_book
DTW construction grids	combined_modified_hoshea, hoshea_zeng/es_only